# Chai Hao Tzun 蔡鎬駿

M.S. Computer Science, National Cheng Kung University

## Background

### Origins & Education

Originally from Sarawak, Malaysia.

Current M.S. in Computer Science at National Cheng Kung University.

## Technical Identity

### Core Focus

Specializing in LLM-based systems.

Experience with web development, DevOps, etc.

Multilingual.

## Personal Interests

### Off the Clock

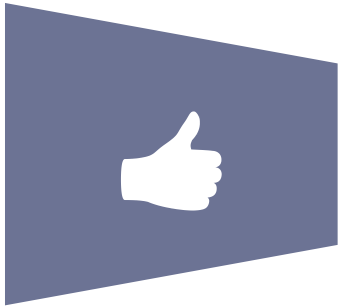
Basketball, bodybuilding, anime, personal finance.

## Mental-RAG

**Skills:**
RAG,
CI/CD,
Web Development,
Collaborative Software Development

- Developed a RAG-based advisory system as part of a lab collaboration with the Ministry of Education
- Goal: equip faculty with professional, case-based guidance for student support
- Automated deployments via Docker & GitLab CI, led RAG benchmarking





## EnHAR-LLM

**Skills:**
LLM,
Research,
Data Visualization

- **Problem: Downsampling wearable sensor data saves battery life but severely degrades activity recognition accuracy**
- **Solution: EnHAR-LLM, a two-stage pipeline that reconstructs missing details from coarse signals, and utilizes a custom alignment adapter to project sensor features directly into the LLM's embedding space**
- **Results: Outperformed existing predecessor, maintaining high accuracy even when data was drastically downsampled**

# When Less Sensing is Enough: Human Activity Recognition from Coarse Motion Signals with Large Language Models

*Haotzun Chai, Ko-Wei Su, Lo Pang-Yun Ting, Kun-Ta Chuang*

13 Mar 2026 ACL 2026 Conference Submission Conference, Haotzun Chai, Ko-Wei Su, Lo Pang-Yun Ting, Kun-Ta Chuang CC BY 4.0

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**Keywords:** Large Language Models, Human Activity Recognition, Sensor Data, Coarse-Grained Data

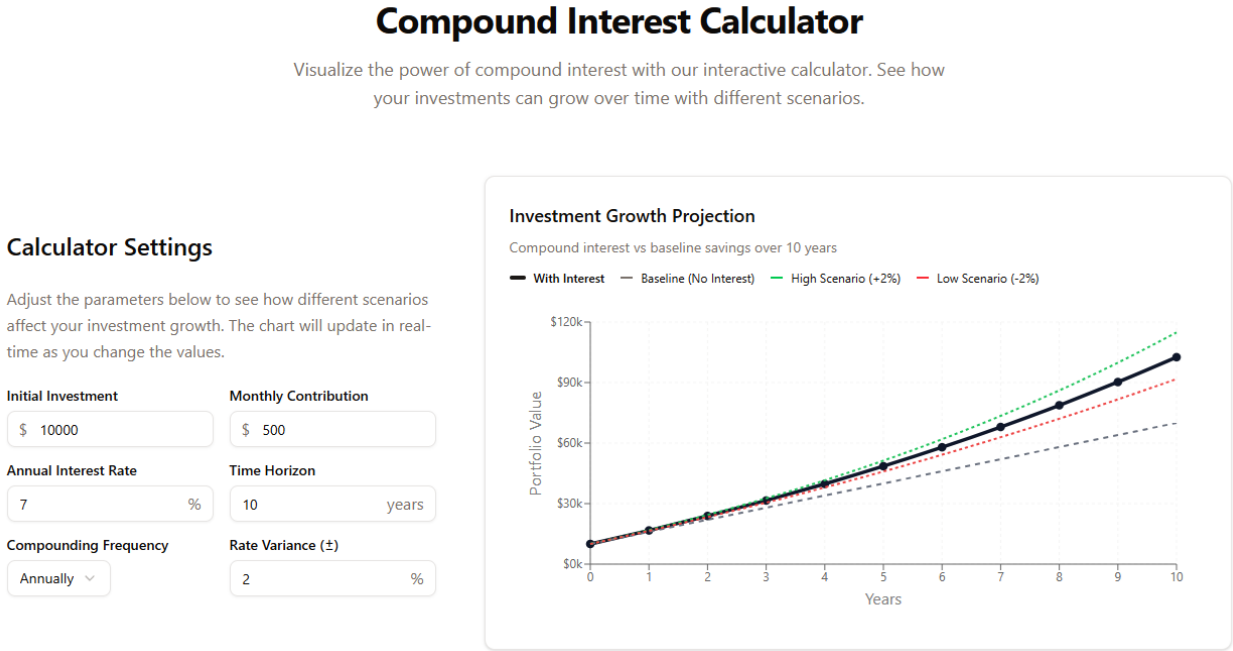
**Abstract:**
Human Activity Recognition (HAR) faces a fundamental trade-off between battery efficiency and sensing precision. While downsampling extends device longevity, the resulting coarse-grained signals often lead to severe information loss and degraded classification accuracy. To address this, we introduce EnHAR-LLM, a novel framework that bridges signal restoration and semantic recognition. First, we utilize an attention-based imputation module to reconstruct high-resolution data while extracting imputation-stage representations as a proxy for signal ambiguity. These features are then integrated via a fusion mechanism into the input space of a frozen Large Language Model (LLM). Extensive experiments on the CAPTURE-24 dataset demonstrate our method achieves robust performance, significantly outperforming baselines even under aggressive downsampling conditions.

**Paper Link:** /forum?id=UQeaPOr5iw
**Response To Metareview:** We sincerely thank the Area Chair, reviewers, and editors for their time and effort in reviewing our work, as well as for recognizing the importance of our problem, model design, and model performance. We are committed to refining our paper by incorporating the valuable feedback and suggestions provided by the reviewers and editors in the next version.
**Research Area:** AI/LLM Agents
**Submission Number:** 5142

## Personal Website

**Skills:**
**Docker,**
**ReactJS,**
**CI/CD,**
**UI Design**

- Consolidated scattered financial tools and a personal portfolio into a single web app for personal use
- Developed a modern, responsive frontend leveraging NextJS
- Containerized the application with Docker, implemented a CI/CD pipeline via GitHub Actions, and hosted on GCP






# Thanks !

Grateful for this opportunity.